

Received 20 April 2026, accepted 12 May 2026, date of publication 25 May 2026, date of current version 5 June 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3696456

RESEARCH ARTICLE

Automated Resume Evaluation Using Large Language Models: A Multi-Agent Framework With Fine-Tuning and Prompt Engineering

MD. SAGOR CHOWDHURY¹, ADIBA FAIROOZ CHOWDHURY¹, AYESHA BANU¹,
RIAD HOSSAIN², AND MAHFUZULHOQ CHOWDHURY¹

¹Department of Computer Science and Engineering, Chittagong University of Engineering and Technology, Chittagong, Bangladesh

²School of Science, Engineering and Technology, East Delta University, Chittagong, Bangladesh

Corresponding author: Riad Hossain (riad.h@eastdelta.edu.bd)

ABSTRACT Automated resume evaluation for assessing candidate-job alignment remains challenging due to the complexity of matching qualifications across education, skills, and experience dimensions. While recent advances in large language models (LLMs) show promise, their effectiveness for domain-specific resume screening remains underexplored. We present a comprehensive benchmark comparing traditional machine learning, deep learning, pre-trained transformers, fine-tuned LLMs, and multi-agent architectures on professionally annotated multi-domain resumes spanning six job categories. Our proposed heterogeneous multi-agent framework—evaluating Education, Skills, and Experience via independently fine-tuned specialist models—achieves the best overall performance (MAE: 0.065, Pearson r : 0.896, R^2 : 0.766; best configuration: Qwen_{FT}+Qwen_{FT}+Gemma_{FT}), outperforming both fine-tuned single-agent models (MAE: 0.064, R^2 : 0.729) and all five commercial state-of-the-art baselines in zero-shot settings: GPT-5 (MAE: 0.252), Gemini 3 (MAE: 0.128), Grok (Auto, xAI) (MAE: 0.121), Mistral Large (MAE: 0.1226), and Claude Sonnet 4.5 (MAE: 0.097). These results show that strategic assignment of domain-specifically fine-tuned models to evaluation sub-dimensions yields interpretable, component-level assessments while achieving superior accuracy over both monolithic and commercial alternatives.

INDEX TERMS Candidate-job alignment, heterogeneous architectures, large language model fine-tuning, multi-agent systems, recruitment automation, resume evaluation.

I. INTRODUCTION

Recruitment and talent acquisition have become increasingly complex in the modern labor market, where organizations receive hundreds or thousands of applications for single positions. Human resource professionals face the challenge of efficiently evaluating candidate qualifications against job requirements while maintaining fairness, consistency, and accuracy. Companies all across the world invest a lot of money in hiring, with the average cost per person being over USD \$4,700 [1]. Furthermore, the manual resume screening takes a lot of time and effort, and the recruiting process can take several weeks [2]. Automated resume

evaluation is becoming not just a convenience but an operational requirement for contemporary hiring systems due to the sheer volume of applications and the requirement for domain-specific expertise to evaluate technical skills.

Even while artificial intelligence has made great strides in natural language comprehension, there are still a number of important obstacles that prevent it from being widely used in resume screening. First, evaluating a CV necessitates careful consideration of a variety of factors, including professional experience, technical proficiency, and educational background, each of which calls for a distinct set of evaluation standards. Second, generalized models are inadequate because candidate credentials and job needs differ significantly across experience levels (entry, mid, and senior) and disciplines (accounting, engineering, and healthcare).

The associate editor coordinating the review of this manuscript and approving it for publication was Jianxiang Xi¹.

Third, the majority of current commercial applicant tracking systems (ATS) focus on rule-based filtering and keyword matching, which are unable to capture contextual relevance, semantic similarity, or the overall quality of candidate profiles [3], [4].

Researchers have put out a variety of computational methods for resume analysis in recent years. While deep learning models like CNN-GRU hybrids and Sentence-BERT have since shown significantly higher performance on candidate ranking and classification tasks [8], [9], early work used TF-IDF-based classifiers and text similarity measures [5], [6], [7]. Large language model (LLM)-based resume screening frameworks, such as single-agent methods [10], [11], and multi-agent pipelines enhanced with retrieval mechanisms [12], have been proposed more recently. Despite these developments, the majority of current research approaches resume evaluation as a coarse ranking or binary classification problem [13], [14]. On a continuous, dimension-specific scoring problem, a thorough and methodical comparison across many paradigms—from conventional machine learning and deep learning to pre-trained transformers, refined LLMs, and multi-agent systems—remains mainly unexplored. Inadequate resume screening has effects that go beyond inefficiencies in operations. Automated systems with poor design have the potential to undermine workforce diversity, disfavor candidates from non-traditional backgrounds, and reinforce hiring biases [4], [15], [16]. Further research has demonstrated that LLM-based screening systems can encode sensitive attribute biases from pre-training data [18] and are susceptible to adversarial manipulation [17], while extensive empirical evidence indicates that, with careful design, AI-assisted pipelines can enhance selection results [19]. These findings underscore the need for transparent, accurate, and dimension-specific evaluation systems that provide both overall alignment scores and interpretable assessments of individual qualification components.

To address these challenges, we present a comprehensive benchmark study comparing multiple approaches to automated resume evaluation using professionally annotated resumes spanning six job domains. Our work evaluates candidate-job alignment across three critical dimensions: education, skills, and experience, with annotations provided by domain experts on a continuous scale (0–1). We systematically compare five modeling paradigms: (a) traditional machine learning with TF-IDF and handcrafted features, (b) deep neural networks including TextCNN, LSTM, BiLSTM, and Transformer architectures, (c) pre-trained transformer models including BERT, RoBERTa, DistilBERT, XLM-RoBERTa, and ALBERT, (d) fine-tuned large language models with optimized LoRA configurations, and (e) multi-agent systems using both homogeneous and heterogeneous model assignments.

Our proposed heterogeneous multi-agent framework, evaluated across 125 distinct model combinations, achieves the best overall results (best configuration:

QwenFT+QwenFT+GemmaFT; MAE = 0.065, Pearson $r = 0.896$, $R^2 = 0.766$), outperforming all single-agent, homogeneous multi-agent, and commercial baselines. Fine-tuned open-source LLMs also substantially outperform commercial models in zero-shot settings—GPT-5 achieves MAE of 0.252, Claude Sonnet 4.5 achieves MAE of 0.097, and Gemini 3 achieves MAE of 0.128—confirming that even leading proprietary systems struggle without domain-specific adaptation. These results show that strategic, fine-tuning-aware model composition is key to accurate and interpretable resume evaluation.

Through this benchmark, we establish methodological foundations for automated resume evaluation that balance performance, interpretability, and computational efficiency. Our work bridges human resource needs with computational capabilities, demonstrating both the potential and current limitations of AI systems for high-stakes hiring decisions.

The main contributions of this work are as follows:

- 1) We establish the first comprehensive five-paradigm benchmark for automated resume evaluation, systematically comparing traditional ML, deep learning, pre-trained transformers, fine-tuned LLMs, and multi-agent architectures under identical experimental conditions.
- 2) We demonstrate that fine-tuned open-source LLMs with optimized LoRA configurations substantially outperform both larger zero-shot models and all five commercial state-of-the-art baselines on domain-specific continuous scoring.
- 3) We systematically evaluate 125 heterogeneous multi-agent configurations and show that strategic assignment of independently fine-tuned specialist models to evaluation dimensions achieves the best overall performance ($R^2 = 0.766$), surpassing single-agent and homogeneous multi-agent configurations.
- 4) We provide a systematic comparison of deep neural network architectures and pre-trained transformer models with and without WordNet-based data augmentation, revealing that augmentation effects are architecture-dependent and non-uniform.
- 5) We apply LIME-based local explainability to each agent's scoring decisions, providing token- and feature-level attribution across all six job domains and three evaluation dimensions to support transparent and auditable hiring.

II. RELATED WORK

A. RESUME DATASETS AND BENCHMARKS

The development of automated resume screening systems has long been constrained by the limited availability of professionally annotated datasets. Most early work relied on proprietary recruitment data or small-scale corpora with binary labels—such as accepted or rejected—without capturing the multi-dimensional nature of candidate-job alignment [5], [6].

A significant step forward came with Heakl et al. [20], who introduced ResumeAtlas, a large-scale dataset of 13,389 resumes from diverse sources and benchmarked large language models, including BERT and Gemma 1.1-2B, achieving a top-1 accuracy of 92%. ResumeAtlas prioritizes classification above ongoing quality rating, even though it increases dataset diversity and scale.

Nevertheless, fine-grained, continuous scores across several qualification variables, including education, abilities, and experience, are still infrequently provided by current datasets.

We present CareerCorpus [21], a professionally annotated multi-domain dataset containing 302 resumes and expert-assigned continuous ratings (0–1) across three evaluation dimensions, in order to overcome this constraint. This is the first benchmark created for dimension-specific alignment score in a regression environment, as far as we are aware.

B. RESUME INFORMATION EXTRACTION

The organized extraction of information from resume documents as a precondition for downstream matching or evaluation has been covered in a sizable body of previous work. Rule-based systems and manually created templates were the foundation of early information extraction techniques, which were later replaced by statistical sequence models that increased robustness across a variety of document formats [22].

The accuracy of parsing was greatly increased with the introduction of pre-trained language models. After being optimized for named entity identification and document classification, transformer-based models like BERT showed excellent performance in a variety of information extraction tasks [23]. Building on these developments, Erdem [25] assessed candidate-job fit for software engineering positions by combining BERT-based named entity identification with fuzzy similarity score and stochastic gradient domain classification. Large-scale structured extraction and relationship mining from resumes have been the subject of more recent research, such as CareerMiner, which uses specialized NER pipelines to build professional networks from resume data [24]. Furthermore, zero-shot resume parsing has been made possible by instruction-tuned big language models, which successfully handle informal and varied resume forms [10].

Assuming access to organized resume sections, our study concentrates on the downstream alignment assessment process rather than extraction itself, even though resume parsing is an essential precondition for any evaluation pipeline.

C. COMPUTATIONAL APPROACHES TO RESUME SCREENING

Automated resume evaluation research has advanced through a number of methodological paradigms. Early methods included classifiers like Naive Bayes, Random Forest, and Support Vector Machines in conjunction with TF-IDF vectorization [5]. For automated applicant shortlisting, the

work in [6] systematically examined Cosine, SqrtCosine, and Improved Sqrt-Cosine (ISC) similarity metrics. They discovered that ISC and Sqrt-Cosine are more closely aligned with expert hiring judgments than regular cosine similarity. To lessen bias and increase candidate matching accuracy, the work in [7] created an ML-based screening system utilizing NLP preprocessing and data-driven criteria. In order to resume classification using tokenization and TF-IDF, the authors in [26] used SVM, logistic regression, random forest, and k-nearest neighbors. SVM achieved the best accuracy, while the work in [27] showed that SVM and random forest above 85% accuracy on IT role categorization. In order to enable real-time multi-category screening, the authors in [28] introduced an AI-powered job matching system that transforms resume text into semantic vectors and rates candidates based on resemblance to job description profiles.

Automatic feature learning was introduced using deep learning architectures. Local ngram patterns are captured by convolutional neural networks, whereas sequential dependencies are modeled by LSTMs and GRUs [14], [29]. For resume categorization and ranking, the work in [8] suggested a CNN-GRU hybrid model that outperformed standalone CNN, LSTM, and conventional ML models with an accuracy of 93.7%.

Sentence-BERT embeddings with cosine similarity scoring were used by the works in [9], who showed 90% accuracy at 0.233 seconds per resume, illustrating the efficacy and efficiency of transformer-based semantic embeddings for candidate ranking.

Large language models have been incorporated into resume evaluation in the most recent round of work. An LLM agent framework for summarizing and grading resumes in bulk was proposed by the works in [11], who reported an 11× gain over manual review and a refined F1 score of 87.73% on sentence categorization. General-purpose LLMs can do competitive zero-shot resume screening given well-structured prompts, but they still perform worse on domain-specific benchmarks than refined smaller models, according to [10]. Despite these advancements, there is still much to learn about the systematic parameter-efficient fine-tuning of open-source LLMs for resume quality regression rather than classification.

D. PERSON-JOB FIT MODELLING

The information retrieval and data mining groups have researched person-job fit (PJF) modeling in great detail, concentrating on the bilateral alignment between job descriptions and candidate profiles [13], [14]. Early PJF systems treated the problem as an asymmetric document retrieval task, representing resumes and job advertisements as bags of words and calculating similarity using cosine distance or BM25. PJF is modeled by more recent neural techniques as a crossattention or co-encoding problem, in which concatenated resume-job pairings are processed by a shared encoder to generate alignment scores [29]. An ability-aware neural network that explicitly predicts the skill gap

between candidate skills and job needs was developed by the authors in [14], who achieved impressive results on extensive industrial recruitment datasets. In a large-scale randomized experiment involving 37,000 applicants, the authors in [19] discovered that AI-assisted recruitment pipelines advanced 54% of candidates through final interviews compared to 34% in traditional pipelines. This finding offers uncommon empirical proof that automated screening can enhance selection quality, despite the fact that AI systems tended to favor younger, less qualified applicants.

The granularity of assessment is a crucial difference between our findings and the larger PJF literature. The majority of PJF systems generate a single overall score without breaking alignment down into independently interpretable sub-dimensions like education, experience, and abilities. By assessing each dimension independently using specialized models and offering component-level diagnostics in addition to an overall score, our multi-agent framework fills this gap.

E. LLM-BASED MULTI-AGENT FRAMEWORKS FOR RECRUITMENT

Multi-agent systems have gained significant traction in recent NLP research. Early ensemble approaches combined homogeneous classifiers via voting or stacking to reduce variance [5], but did not exploit complementary strengths arising from architectural diversity.

The recent proliferation of LLMs has renewed interest in heterogeneous agent architectures. Gan et al. [11] demonstrated that LLM agents can be used for decision-making in the final offer stage of recruitment, showing promising automation of late-stage hiring steps. Lo et al. [12] proposed a four-agent framework—resume extractor, evaluator, summariser, and score formatter—augmented with Retrieval-Augmented Generation (RAG) to incorporate industry-specific knowledge such as university rankings and certification databases, with scores validated against HR professional ratings. While both frameworks advance multi-agent recruitment automation, neither investigates the effect of heterogeneous model assignment across evaluation dimensions, nor do they conduct a systematic search over model combinations. Bian et al. [13] employed a multi-view co-teaching architecture for person–job matching, implicitly decomposing the alignment problem across multiple views, but without assigning different model architectures to different roles.

Our work directly addresses this gap by conducting a systematic grid search over 125 heterogeneous multi-agent configurations, characterising when heterogeneous fine-tuned model assignment outperforms both homogeneous multi-agent and single-agent baselines.

F. PARAMETER-EFFICIENT FINE-TUNING OF LARGE LANGUAGE MODELS

Fine-tuning large language models for downstream tasks has evolved significantly since the introduction of full-parameter

supervised fine-tuning [30], [31]. Full fine-tuning becomes computationally prohibitive for billion-parameter models and is prone to catastrophic forgetting on limited datasets.

Parameter-efficient fine-tuning (PEFT) methods address these limitations by updating only a small fraction of parameters. Low-Rank Adaptation (LoRA) [32] has emerged as the dominant PEFT approach, decomposing weight updates into low-rank matrix products to dramatically reduce trainable parameters while preserving task-specific expressivity. QLoRA further extends these efficiency gains by combining LoRA with 4-bit quantised base weights, enabling fine-tuning of 10B+ parameter models on consumer hardware [33]. Mu et al. [17] demonstrated that LoRA-based adaptation can also be applied defensively—their FIDS method uses LoRA to detect adversarial instructions injected into resumes, achieving 15.4% attack reduction with only 10.4% false rejection increase.

Our work applies LoRA fine-tuning to four open-source models for resume quality regression, among the first systematic comparisons of LoRA-fine-tuned LLMs on this task.

G. BIAS, FAIRNESS, AND SECURITY IN AUTOMATED HIRING

The potential for automated employment systems to encode and magnify societal biases has drawn a lot of attention. When Raghavan et al. [4] looked at a variety of algorithmic hiring tools, they discovered that many of them reinforce historical inequities related to gender, ethnicity, and educational background. Even in the absence of explicit gender cues, algorithmic ad targeting in STEM recruitment appears to discriminate against women, as reference [15] empirically showed. In their thorough analysis of bias types, fairness indicators, and mitigation strategies in AI-driven hiring, the authors in [16] point out important gaps and offer solutions for more equal recruiting procedures.

These issues have been addressed at the model level in more recent work. In order to achieve higher fairness metrics while preserving competitive accuracy, the work in [18] devised an adversarial contrastive tuning approach that uses adapter-based PEFT to separate qualification signals from sensitive attribute signals in LLM-based resume screening. The authors in [17] have found that LLMs used for resume screening are susceptible to adversarial instructions concealed in resume text, with attack success rates over 80% for several attack types, highlighting the fact that security is a crucial practical concern in addition to fairness.

Our framework reduces keyword bias by semantic scoring and produces comprehensible component-level outputs that facilitate human auditing, even though a thorough bias and adversarial analysis is outside the purview of this study. Future research should incorporate adversarial robustness methods and fairness restrictions into the fine-tuning pipeline.

H. RESEARCH GAPS

Despite the substantial progress reviewed above, several critical gaps remain. First, there is no systematic benchmark comparing traditional ML, deep learning, pre-trained transformers, and fine-tuned LLMs under identical conditions for *continuous* resume alignment scoring, as opposed to classification or ranking. Second, parameter-efficient fine-tuning of open-source LLMs specifically for resume quality regression remains underexplored; most LLM-based work uses zero-shot or few-shot prompting. Third, heterogeneous multi-agent architectures that assign different fine-tuned models to different evaluation dimensions have received no attention; existing multi-agent recruitment systems either use homogeneous model pools or rely on RAG augmentation. Fourth, data augmentation strategies for low-resource LLM fine-tuning in the recruitment domain remain largely unexplored. Fifth, explainability of automated scoring decisions—essential for fair and transparent hiring—has not been addressed in the context of LLM-based resume evaluation.

Our work addresses all five gaps through a comprehensive, five-paradigm benchmark with 125 heterogeneous multi-agent configurations and LIME-based explainability analysis across all evaluation dimensions and job domains.

III. METHODOLOGY

Fig. 1 illustrates the system architecture of the proposed resume evaluation framework, outlining the end-to-end process from model preparation to performance benchmarking against state-of-the-art systems. The pipeline consists of two primary stages: Stage 1 focuses on model preparation through data augmentation, fine-tuning, and prompt engineering, while Stage 2 implements both single-agent and multi-agent evaluation strategies with comprehensive performance comparison against commercial large language models.

A. TASK DEFINITION

The three organized elements of a resume are education (E_i), skills (K_i), and professional experience (X_i). The task is presented as a regression issue where resume content is mapped to a continuous quality score using a scoring function $f : r_i \rightarrow [0, 1]$. Stronger congruence between a candidate's qualifications and the particular criteria of a certain job area (such as research assistant or accountant) at a particular seniority level (entry-level, mid-level, and senior-level) is indicated by higher values.

Evaluation is divided into three specialized subtasks for the multi-agent architecture, each of which is managed by a dedicated agent. The Education Agent ($f_E: E_i \rightarrow [0, 1]$) evaluates field alignment, GPA, institutional excellence, and degree relevance. The Skills Agent ($f_K: K_i \rightarrow [0, 1]$) assesses tool proficiency, certifications, and technical competencies. Work history, role growth, and achievement significance are all analyzed by the Experience Agent ($f_X: X_i \rightarrow [0, 1]$). The arithmetic mean of the three component scores determines the

final aggregated score:

$$\hat{s}_i = \frac{1}{3} (f_E(E_i) + f_K(K_i) + f_X(X_i)) \quad (1)$$

B. DATASET

We used the CareerCorpus dataset [21], which includes 302 professionally annotated resumes from three seniority levels (Entrylevel, Mid-level, and Senior-level) and six employment fields (Accountant, Apparel, Banking, Finance, Research Assistant, and Teacher). Two professional annotators separately graded each resume on a continuous scale ranging from 0.0 to 1.0, resulting in an inter-annotator agreement of $r = 0.78$ (Pearson correlation, $p < 0.001$).

The dataset shows an unequal distribution across seniority levels and job domains, as seen in Fig. 2. Resumes from research assistants and accountants make up the majority of samples, whilst those from the apparel and banking industries are relatively underrepresented. Senior-level resumes make up the majority across all seniority levels, while entrance-level and mid-level examples are less common. In order to maintain domain and seniority proportions in both the training and test splits, we employed stratified sampling for dataset splitting in order to address this imbalance.

Thus, stratified sampling was used to divide the dataset into a training set (80%, 258 resumes) and a test set (20%, 44 resumes). The non-augmented training setup employed all 258 training samples directly due to the restricted amount of training data, but the augmented configuration quadrupled this to 516 samples by synonym replacement, as explained in Section III-C. The complete distribution statistics are shown in Fig. 2.

C. DATA AUGMENTATION

Lexical-level text augmentation was used to increase model resilience and reduce overfitting given the small training set of 258 examples. In particular, each resume's Education, Skills and Achievements, and Experience sections were subjected to WordNet-based synonym substitution using the NLTK library at a predetermined replacement rate of thirty percent. To maintain grammatical accuracy and semantic consistency, short words and non-alphabetic tokens were eliminated. The training set was effectively doubled from 258 to 516 samples but label consistency was maintained because each original resume generated one enhanced variant that retained the original quality score. After that, an 85–15 split was used to divide the augmented set into 438 training and 78 validation samples.

For low-resource natural language processing jobs, this adheres to typical data augmentation techniques [36].

D. MODEL PREPARATION: STAGE 1

1) BASE MODEL SELECTION

Based on their success in numerical reasoning and their ability to follow instructions, four cutting-edge open-source big language models were chosen. A 14 billion parameter model with robust analytical logic is Qwen 2.5-14B [30].

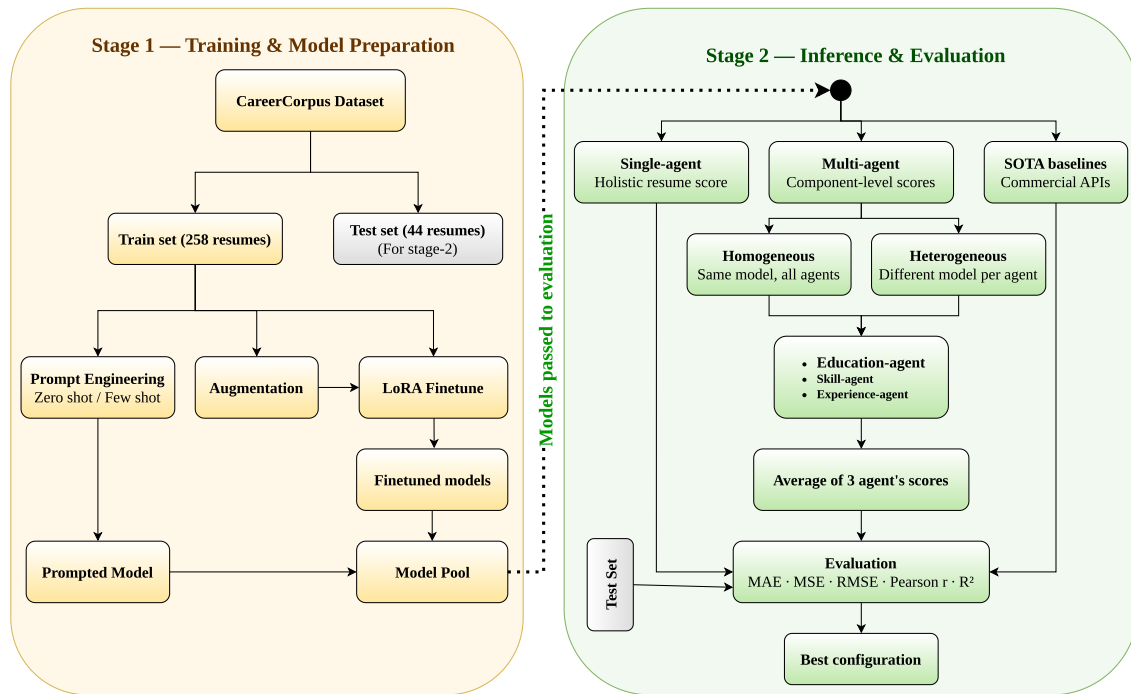


FIGURE 1. Overall pipeline for automated resume evaluation system.

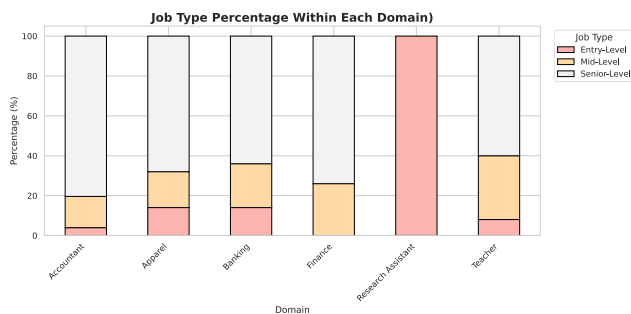


FIGURE 2. CareerCorpus dataset distribution.

A 22 billion parameter model called Mistral Small 22B [34] has been optimized for challenging evaluation tasks. Google created the effective 7 billion parameter model known as Gemma 7B [35]. Meta’s 8 billion parameter model, Llama 3.1-8B [31], is valued for its wide task adaptability. This selection offers a wide variety of architectural designs and model capacities appropriate for methodical comparison examination.

2) FINE-TUNING WITH LORA

Low-Rank Adaptation (LoRA) [32], a parameter-efficient method that adds trainable low-rank matrices to the frozen base model weights, was used to fine-tune all models. The LoRA configuration targeted the Query, Key, Value, and Output projection layers and used a rank of $r = 32$, a scaling factor of $\alpha = 64$ (maintaining a 2:1 ratio with the rank), and

a dropout rate of 0.05. The AdamW optimizer was used for training at a learning rate of 3×10^{-4} with cosine annealing, an effective batch size of 16 (physical batch size of 2 with gradient accumulation over 8 steps), a warmup phase that covered 10% of all training steps, a weight decay of 0.01, a maximum sequence length of 768 tokens, and mixed FP16 precision over 5 epochs. A nonaugmented variation trained on all 258 original training samples and an augmented variant trained on 516 samples, including synthetic variants created using the process outlined in Section III-C, were the two training configurations that were assessed.

3) PROMPT ENGINEERING

For non-fine-tuned evaluation, two complementary prompting strategies were implemented. Under *zero-shot prompting*, models received detailed task instructions, evaluation criteria, and resume content without any example demonstrations. Each prompt included the task definition with the scoring scale (0.0–1.0), component-wise evaluation criteria covering education, skills, experience, and projects, contextual metadata on job domain and seniority level, and a strict output format specification requiring a numerical score only.

Under *few-shot prompting*, each model was provided with five annotated example resumes prior to evaluating the target resume. Examples were stratified across three quality tiers—low (0.2–0.4), medium (0.5–0.7), and high (0.8–1.0)—to supply calibrated demonstrations of the scoring process across the full score range.

E. EVALUATION FRAMEWORK: STAGE 2

1) SINGLE-AGENT EVALUATION

In the single-agent configuration, a model processes the complete resume holistically and produces a single quality score. The model receives structured resume content encompassing education, skills and experience alongside job domain and seniority metadata, and outputs a prediction $\hat{s}_i \in [0, 1]$. This architecture serves as the primary baseline against which the benefit of multi-agent decomposition is measured.

2) MULTI-AGENT EVALUATION

The multi-agent framework assigns each resume component to a dedicated expert agent, each receiving a component-specific prompt tailored to its evaluation focus. The *Education Agent* evaluates degree type, field relevance, institutional prestige, and GPA. The *Skills Agent* assesses technical competencies, programming languages, tools, frameworks, certifications and achievements. The *Experience Agent* analyses work history duration, role progression, and domain relevance. Each agent independently produces a component score in $[0, 1]$, and the three scores are aggregated via simple arithmetic mean as defined in (1).

Two multi-agent variants were evaluated. In the *homogeneous* configuration, all three agents employ the same underlying model, ensuring consistency in reasoning patterns while leveraging component-specific prompting. In the *heterogeneous* configuration, different models are assigned to different agents according to their empirical strengths; for example, the top-performing configuration assigns fine-tuned Qwen 2.5-14B to the Education and Skills agents and fine-tuned Gemma 7B to the Experience agent. The relative merits of these two configurations are examined in Section IV.

3) STATE-OF-THE-ART BASELINE COMPARISON

To establish competitive benchmarks, all five commercial models were accessed via their official web interfaces — Claude Sonnet 4.5 (claude.ai), GPT-5 (chatgpt.com), Gemini 3 (gemini.google.com), Grok (grok.com), and Mistral Large (chat.mistral.ai) — using identical zero-shot prompts. Web interfaces were used due to API access constraints at the time of evaluation; outputs reflect default generation settings. To minimise variability, each resume was evaluated in a fresh session, and evaluations were conducted within a single 48-hour window.

F. EVALUATION METRICS

Model performance was assessed using five standard regression metrics, each computed separately against Annotator-1 scores, Annotator-2 scores, and the average annotator score as the primary evaluation target. All metrics were evaluated on the held-out test set.

Mean Absolute Error (MAE) measures the average absolute deviation between predicted and ground-truth

scores:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |s_i - \hat{s}_i| \quad (2)$$

Mean Squared Error (MSE) penalises larger deviations more heavily by squaring individual errors:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (s_i - \hat{s}_i)^2 \quad (3)$$

Root Mean Square Error (RMSE) restores the error to the original scale of the target variable:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (s_i - \hat{s}_i)^2} \quad (4)$$

Pearson Correlation Coefficient (r) quantifies the linear association between predicted and ground-truth scores:

$$r = \frac{\sum_{i=1}^N (s_i - \bar{s})(\hat{s}_i - \bar{\hat{s}})}{\sqrt{\sum_{i=1}^N (s_i - \bar{s})^2} \sqrt{\sum_{i=1}^N (\hat{s}_i - \bar{\hat{s}})^2}} \quad (5)$$

Coefficient of Determination (R^2) captures the proportion of variance in annotator scores explained by the model:

$$R^2 = 1 - \frac{\sum_{i=1}^N (s_i - \hat{s}_i)^2}{\sum_{i=1}^N (s_i - \bar{s})^2} \quad (6)$$

where N is the size of the test set, s_i is the ground-truth annotator score, $\hat{s}_i$ is the model prediction, and $\bar{s}$ and $\bar{\hat{s}}$ are the corresponding sample means. While greater values of r and R^2 show a stronger correlation with human judgments, lower values of MAE, MSE, and RMSE suggest superior predictive accuracy.

G. EXPERIMENTAL CONFIGURATIONS

Several orthogonal dimensions were included in the experimental evaluation. Models were trained using both augmented (516 samples) and non-augmented (258 samples) training strategies. For inference approach, three modes were considered: zero-shot prompting, few-shot prompting with five stratified demonstrations, and fine-tuned inference. Regarding evaluation architecture, three setups were examined: single-agent holistic evaluation, homogeneous multi-agent evaluation employing the same model for all components, and heterogeneous multi-agent evaluation assigning different models per component. Across all combinations of the four base models, training strategies, inference approaches, and evaluation architectures, more than 40 distinct configurations were systematically evaluated

against the held-out test set, with performance reported across all metrics and evaluation targets (Annotator-1, Annotator-2, and Average).

IV. RESULTS

This section presents a systematic evaluation of all model configurations against the held-out test set. Results are reported primarily against the average annotator score (hereafter “Avg”), with supplementary breakdowns for Annotator-1 and Annotator-2 where instructive. We organise the discussion into nine parts: traditional machine learning baselines (Section IV-A), deep neural network evaluation (Section IV-B), pre-trained transformer evaluation (Section IV-C), single-agent LLM evaluation (Section IV-D), homogeneous multi-agent evaluation (Section IV-E), heterogeneous multi-agent evaluation (Section IV-F), comparison with commercial state-of-the-art baselines (Section IV-G), LIME-based explainability analysis (Section IV-J), and computational complexity analysis (Section IV-K).

A. TRADITIONAL MACHINE LEARNING BASELINES

Table 1 reports the performance of six classical machine learning models—Ridge Regression, Lasso Regression, ElasticNet, Random Forest, Gradient Boosting, and XGBoost—alongside a Support Vector Regressor (SVR), all trained on TF-IDF features extracted from resume text. Results are reported across all three evaluation targets.

Among linear models, Lasso Regression achieved the strongest performance, attaining $R^2 = 0.5752$ and MAE = 0.0870 against the average annotator score, outperforming both Ridge ($R^2 = 0.3354$, MAE = 0.1028) and ElasticNet ($R^2 = 0.4382$, MAE = 0.1008). The superior performance of Lasso over Ridge and ElasticNet suggests that sparse feature selection is beneficial for resume text representation, as many TF-IDF dimensions are uninformative for quality prediction.

Linear regressors were regularly outperformed by ensemble tree-based models. With an R^2 of 0.6329 (MAE = 0.0736, Pearson $r = 0.8369$) against the average annotator score, XGBoost outperformed all other conventional ML models. Random Forest followed closely with $R^2 = 0.6452$ (MAE = 0.0730, $r = 0.8340$), and Gradient Boosting achieved $R^2 = 0.6207$ (MAE = 0.0764, $r = 0.8089$). Despite having great rank-ordering abilities, SVR produced competitive correlation ($r = 0.8618$), lesser $R^2 = 0.4821$, and somewhat high MAE = 0.1055, indicating systematic bias in its absolute score predictions.

All models show the same pattern: performance against Annotator-1 scores consistently outperforms performance against Annotator-2, and the average-target outcomes outperform both. This pattern, which is also seen in LLM-based settings, indicates that by lowering individual annotator noise, the average annotator score offers a smoother, more learnable regression goal.

Random Forest and XGBoost produce the best results ($R^2 \approx 0.63$ – 0.64) out of all the typical ML models, proving that well calibrated hand-crafted feature pipelines

TABLE 1. Traditional ML model performance across all three evaluation targets. Best result per metric (Average target) is highlighted in bold.

Model	Target	R^2	Pearson r	MSE	RMSE	MAE
Ridge	Ann.-1	0.3373	0.7122	0.0249	0.1579	0.1238
	Ann.-2	0.1213	0.7211	0.0215	0.1468	0.1168
	Avg	0.3354	0.7615	0.0181	0.1344	0.1028
Lasso	Ann.-1	0.5940	0.7728	0.0153	0.1236	0.0919
	Ann.-2	0.2590	0.7252	0.0182	0.1348	0.1082
	Avg	0.5752	0.7991	0.0116	0.1075	0.0870
ElasticNet	Ann.-1	0.4795	0.7327	0.0196	0.1400	0.1109
	Ann.-2	0.1310	0.6909	0.0213	0.1460	0.1142
	Avg	0.4382	0.7592	0.0153	0.1236	0.1008
Random Forest	Ann.-1	0.6318	0.8057	0.0139	0.1177	0.0830
	Ann.-2	0.3563	0.7580	0.0158	0.1256	0.0977
	Avg	0.6452	0.8340	0.0096	0.0982	0.0730
Gradient Boosting	Ann.-1	0.5989	0.7923	0.0151	0.1229	0.0929
	Ann.-2	0.3524	0.7218	0.0159	0.1260	0.0982
	Avg	0.6207	0.8089	0.0103	0.1015	0.0764
XGBoost	Ann.-1	0.6437	0.8163	0.0134	0.1158	0.0830
	Ann.-2	0.3106	0.7510	0.0169	0.1300	0.1028
	Avg	0.6329	0.8369	0.0100	0.0999	0.0736
SVR	Ann.-1	0.5271	0.8274	0.0178	0.1334	0.1097
	Ann.-2	0.1553	0.7898	0.0207	0.1439	0.1279
	Avg	0.4821	0.8618	0.0141	0.1187	0.1055

TABLE 2. Deep neural network performance on the average annotator target, with and without WordNet augmentation. Best result per condition is in bold.

Model	Aug.	MAE	MSE	RMSE	Pearson r	R^2
TextCNN	No	0.0778	0.0102	0.1010	0.8399	0.6246
	Yes	0.0776	0.0110	0.1050	0.8634	0.5947
LSTM	No	0.0936	0.0156	0.1248	0.6655	0.4275
	Yes	0.1019	0.0203	0.1425	0.5625	0.2527
BiLSTM	No	0.0797	0.0120	0.1097	0.8622	0.5573
	Yes	0.0713	0.0103	0.1013	0.8405	0.6229
Transformer	No	0.0944	0.0187	0.1369	0.8314	0.3110
	Yes	0.0861	0.0156	0.1250	0.8528	0.4250

might be a useful lower bound for resume quality prediction. These findings set the standard that later deep learning and LLM-based methods are anticipated to surpass.

B. DEEP NEURAL NETWORK EVALUATION

Table 2 reports performance of the four deep neural network architectures against the average annotator score under both non-augmented and augmented training conditions.

Without augmentation, TextCNN achieved the strongest result ($R^2 = 0.6246$, MAE = 0.0778, Pearson $r = 0.8399$), closely followed by BiLSTM ($R^2 = 0.5573$, MAE = 0.0797, $r = 0.8622$). LSTM and the custom Transformer both lagged behind within the group, attaining $R^2 = 0.4275$ and $R^2 = 0.3110$ respectively, with the Transformer showing the highest Pearson r of its group at 0.8314 despite its lower R^2 , indicating systematic score bias rather than poor rank-ordering.

TABLE 3. Pre-trained transformer model performance on the average annotator target. ΔR^2 reports the change from no-augmentation to augmentation (positive = improvement). Best no-aug and aug results are individually bolded.

Model	Aug.	MAE	MSE	RMSE	Pearson r	R^2	ΔR^2
BERT-base	No	0.0758	0.0100	0.1002	0.8272	0.6306	-0.0454
	Yes	0.0668	0.0113	0.1062	0.8591	0.5852	
RoBERTa-base	No	0.0701	0.0095	0.0976	0.8480	0.6495	+0.0598
	Yes	0.0647	0.0079	0.0889	0.8568	0.7093	
DistilBERT	No	0.0588	0.0068	0.0825	0.8726	0.7495	-0.0678
	Yes	0.0627	0.0087	0.0930	0.8481	0.6817	
XLM-RoBERTa	No	0.1092	0.0222	0.1489	0.7858	0.1847	-0.0868
	Yes	0.1254	0.0245	0.1566	0.6423	0.0979	
ALBERT-base	No	0.0656	0.0070	0.0838	0.8635	0.7420	-0.0142
	Yes	0.0661	0.0074	0.0860	0.8834	0.7278	

Data augmentation produced mixed results across architectures. BiLSTM benefited most, improving from $R^2 = 0.5573$ to 0.6229 and reducing MAE from 0.0797 to 0.0713 , making it the best-performing deep learning model under augmented training. TextCNN showed only marginal improvement ($R^2: 0.6246 \rightarrow 0.5947$), while LSTM degraded noticeably ($R^2: 0.4275 \rightarrow 0.2527$). The custom Transformer improved moderately ($R^2: 0.3110 \rightarrow 0.4250$). The selective benefit of augmentation for BiLSTM suggests that bidirectional sequential models generalise better from synonym-enriched training data, whereas LSTM is more sensitive to the distributional shift introduced by augmentation.

C. PRE-TRAINED TRANSFORMER EVALUATION

Table 3 reports the performance of five pre-trained transformer models fine-tuned for resume quality regression, evaluated without and with WordNet synonym augmentation against the average annotator score. Augmentation impact (Δ) is reported as the difference $\text{aug} - \text{no aug}$ for R^2 and MAE.

Without augmentation, DistilBERT achieved the highest R^2 among the pre-trained models at $R^2 = 0.7495$ (MAE = 0.0588 , Pearson $r = 0.8726$). ALBERT-base followed with $R^2 = 0.7420$ (MAE = 0.0656 , $r = 0.8635$), while RoBERTa-base achieved $R^2 = 0.6495$ and BERT-base reached $R^2 = 0.6306$. XLM-RoBERTa-base performed substantially below the other models at $R^2 = 0.1847$ (MAE = 0.1092), likely because its multilingual pretraining is misaligned with the English-only CareerCorpus domain.

The effect of augmentation is heterogeneous across models. RoBERTa-base showed the largest gain, improving by $\Delta R^2 = +0.0598$ (MAE: $0.0701 \rightarrow 0.0647$). ALBERT-base showed a modest decline ($\Delta R^2 = -0.0142$), while DistilBERT degraded more noticeably ($\Delta R^2 = -0.0678$), suggesting that its distilled architecture is susceptible to overfitting on noisier synonym-substituted samples. BERT-base also declined under augmentation ($\Delta R^2 = -0.0454$). XLM-RoBERTa-base degraded further with augmentation ($\Delta R^2 = -0.0868$). Across all five models, DistilBERT

without augmentation and RoBERTa-base with augmentation represent the strongest results within the pre-trained transformer group.

D. SINGLE-AGENT LLM EVALUATION

1) EFFECT OF FINE-TUNING AND DATA AUGMENTATION

Table 4 summarises single-agent performance across all four base models under zero-shot, few-shot, non-augmented fine-tuning, and augmented fine-tuning conditions. Without fine-tuning, all models exhibit moderate predictive accuracy. Few-shot prompting consistently outperformed zero-shot prompting across all models. Qwen 2.5-14B under few-shot conditions achieved MAE = 0.0976 , Pearson $r = 0.6767$, and $R^2 = 0.3187$ against the average annotator score, compared to MAE = 0.1083 and $r = 0.4022$ under zero-shot conditions. Mistral Small 22B (few-shot) reached MAE = 0.0824 and $r = 0.6856$, representing the strongest prompting-only configuration. Llama 3.1-8B (few-shot) achieved MAE = 0.0913 and $r = 0.6967$, closely competitive with the larger models, while the Qwen3-14B zero-shot baseline yielded $r = 0.2980$, highlighting the substantial gap between zero-shot and calibrated prompting. GPT-OSS-20B in zero-shot conditions achieved MAE = 0.0908 and $r = 0.4297$ ($R^2 = 0.0295$), with few-shot prompting improving correlation to $r = 0.6185$ ($R^2 = 0.2648$), confirming that few-shot calibration benefits open-weight models as well as the open-source alternatives evaluated above. Qwen3-14B zero-shot yielded the weakest prompting result ($r = 0.2980$, $R^2 = -0.5549$), though few-shot prompting recovered performance substantially to $r = 0.6370$ ($R^2 = 0.2243$), suggesting Qwen3-14B requires in-context demonstrations to calibrate its numerical output range for this task. Additionally, two open-weight models were evaluated in zero-shot and few-shot settings only: Qwen3-14B, the instruction-tuned successor to Qwen 2.5-14B, and GPT-OSS-20B [38], OpenAI's open-weight model released in August 2025. Neither was included in fine-tuning experiments due to computational constraints.

All models' performance was consistently harmed by fine-tuning without data replenishment. Without augmentation, Qwen 2.5-14B yielded $R^2 = -0.1101$ (MAE = 0.1420), Gemma 7B yielded $R^2 = -1.8565$ (MAE = 0.2307), and Mistral Small 22B yielded $R^2 = -2.1822$ (MAE = 0.2447), with models collapsing to nearly constant predictions. Without augmentation, Llama 3.1-8B suffered the most, with $R^2 = -8.1493$ (MAE = 0.4495). This suggests that serious overfitting and score-range collapse result with fine-tuning on just 258 samples without augmentation.

For the majority of models, data augmentation significantly corrected this decline. The best single-agent result was obtained by Qwen 2.5-14B with augmentation and the revised LoRA configuration (rank 32, $\alpha = 64$, 5 epochs, cosine schedule): MAE = 0.0644 , RMSE = 0.0858 , Pearson $r = 0.8541$, and $R^2 = 0.7294$ versus the average annotator score. This is a significant improvement over the model's

TABLE 4. Single-agent evaluation results on the average annotator score. Best fine-tuned result is highlighted in bold.

Model	Configuration	MAE	MSE	RMSE	Pearson r	R^2
Qwen 2.5-14B	Zero-shot	0.1083	0.0246	0.1569	0.4022	0.0957
	Few-shot	0.0976	0.0185	0.1361	0.6767	0.3187
	FT (no aug.)	0.1420	0.0302	0.1737	0.7117	-0.1101
	FT (aug.)	0.0644	0.0074	0.0858	0.8541	0.7294
Qwen3-14B	Zero-shot	0.1338	0.0423	0.2056	0.2980	-0.5549
	Few-shot	0.0978	0.0211	0.1453	0.6370	0.2243
Mistral Small 22B	Zero-shot	0.1581	0.0463	0.2151	0.2742	-0.7017
	Few-shot	0.0824	0.0169	0.1302	0.6856	0.3768
	FT (no aug.)	0.2447	0.0865	0.2941	-0.1078	-2.1822
	FT (aug.)	0.1301	0.0419	0.2046	-0.1716	-0.5402
Gemma 7B	FT (no aug.)	0.2307	0.0777	0.2787	0.3370	-1.8565
	FT (aug.)	0.2088	0.0556	0.2357	0.4337	-1.0434
Llama 3.1-8B	Few-shot	0.0913	0.0168	0.1297	0.6967	0.3241
	FT (no aug.)	0.4495	0.2487	0.4987	0.3672	-8.1493
	FT (aug.)	0.4532	0.2689	0.5185	0.3979	-8.8910
GPT-OSS-20B	Zero-shot	0.0908	0.0264	0.1624	0.4297	0.0295
	Few-shot	0.0917	0.0200	0.1414	0.6185	0.2648

FT(no aug)=Fine-tuned (no augmentation), FT(aug)=Fine-tuned (augmented)

TABLE 5. Homogeneous multi-agent configurations ranked by R^2 against the average annotator score. All three agents use the same model.

Model (all agents)	MAE	MSE	RMSE	Pearson r	R^2
Qwen 2.5-14B (FT)	0.0775	0.0092	0.0959	0.8736	0.6619
Mistral Small 22B (FT)	0.0821	0.0106	0.1032	0.7923	0.6084
Gemma 7B (FT)	0.0943	0.0168	0.1297	0.7681	0.3816
Mistral Small 22B (pre)	0.1101	0.0173	0.1315	0.7394	0.3644
Llama 3.1-8B (pre)	0.1485	0.0289	0.1700	0.7727	-0.0633

(FT) = Fine-tuned; (pre) = pre-trained zero-shot

non-augmented version. Mistral Small 22B with augmentation improved to MAE = 0.1301, but R^2 remained negative (-0.5402), indicating that the 22B model requires more training data than the augmented set provides to achieve reliable score calibration. Gemma 7B with augmentation similarly failed to improve beyond negative R^2 (-1.0434), suggesting that its 7B capacity is insufficient to generalise from limited augmented data alone. Llama 3.1-8B showed no meaningful benefit from augmentation, with R^2 = -8.8910, and is not a suitable choice for fine-tuned single-agent evaluation on this dataset.

E. HOMOGENEOUS MULTI-AGENT EVALUATION

Table 5 reports aggregated performance for all five homogeneous multi-agent configurations, in which the same model is assigned to all three agents (Education, Skills, and Experience), evaluated against the average annotator score.

The best homogeneous configuration is Qwen 2.5-14B (fine-tuned), achieving MAE = 0.0775, RMSE = 0.0959, Pearson r = 0.8736, and R^2 = 0.6619. Fine-tuned Mistral Small 22B followed with R^2 = 0.6084 (MAE = 0.0821), demonstrating that fine-tuned models consistently outperform their pre-trained counterparts in the homogeneous setting. Gemma 7B fine-tuned achieved R^2 = 0.3816 (MAE = 0.0943), while pre-trained Mistral Small 22B yielded R^2 = 0.3644 (MAE = 0.1101). Pre-trained Llama 3.1-8B produced the weakest result with R^2 = -0.0633 (MAE = 0.1485),

indicating systematic score miscalibration despite a reasonable Pearson r of 0.7727.

Notably, the best homogeneous result (R^2 = 0.6619, Qwen all-FT) remains below both the best single-agent fine-tuned result (R^2 = 0.7294) and the best heterogeneous configuration (R^2 = 0.7664), suggesting that simple arithmetic aggregation of component scores introduces noise that offsets the gains from component-specific prompting. Component-level scores nonetheless provide interpretable diagnostic value for practical resume screening systems.

F. HETEROGENEOUS MULTI-AGENT EVALUATION

To assess whether assigning specialised models to individual agents could improve upon homogeneous designs, we evaluated all 125 heterogeneous configurations (5^3 combinations of five model variants across three agent roles) drawn from the available pool: Qwen 2.5-14B fine-tuned, Mistral Small 22B fine-tuned, Mistral Small 22B pre-trained, Gemma 7B fine-tuned, and Llama 3.1-8B pre-trained, assigned across the Education, Skills, and Experience agent roles. Each configuration was evaluated on the held-out test set and ranked by R^2 against the average annotator score.

Table 6 reports the top-ten configurations. The best heterogeneous configuration—fine-tuned Qwen 2.5-14B for the Education Agent, fine-tuned Qwen 2.5-14B for the Skills Agent, and fine-tuned Gemma 7B for the Experience Agent—achieved MAE = 0.0654, RMSE = 0.0797, Pearson r = 0.8959, and R^2 = 0.7664 against the average annotator score, outperforming both the best homogeneous multi-agent result (R^2 = 0.6619) and the best single-agent fine-tuned configuration (R^2 = 0.7294). Fig. 3 illustrates the inference pipeline of this best-found configuration.

1) ARCHITECTURE

The three agents operate in parallel, each receiving a dedicated component-specific input. The Education Agent receives the *Education* section of the resume, the Skills

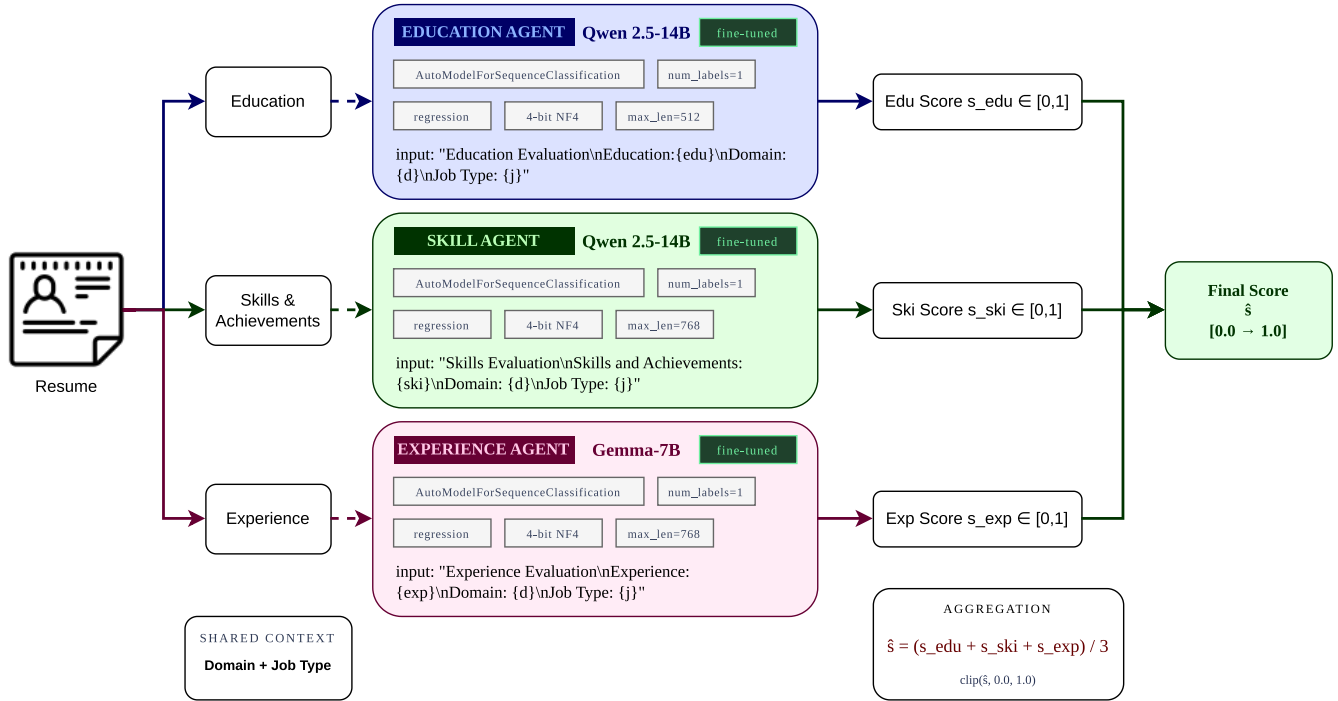


FIGURE 3. Inference pipeline of the best-found heterogeneous multi-agent configuration for automated resume evaluation.

Agent receives the *Skills and Achievements* section, and the Experience Agent receives the *Experience* section. All agents additionally receive the resume's domain and job-type metadata as shared context. Each agent independently produces a component score $\hat{s} \in [0, 1]$ using an `AutoModelForSequenceClassification` head with `num_labels=1` and `problem_type=regression`. The three scores are aggregated via arithmetic mean and clipped to $[0, 1]$ as defined in (1).

2) MODEL AND TRAINING DIFFERENCES

Although both the Education and Skills agents employ the same base model (Qwen 2.5-14B), they were fine-tuned independently on their respective resume sections with section-specific data augmentation. For the Education Agent, synonym substitution was applied exclusively to the Education column; for the Skills Agent, augmentation targeted the Skills and Achievements column; and for the Experience Agent (Gemma 7B), augmentation was applied to the Experience column. This ensures each agent's parameters are specialised for its evaluation focus rather than generalised across sections.

Key training differences between agents are worth noting. The Qwen-based Education and Skills agents targeted seven LoRA modules—query, key, value, output, gate, up, and down projections—with learning rate 3×10^{-4} and maximum sequence length 512 and 768 tokens respectively, reflecting the typically shorter length of education descriptions versus skills listings. The Gemma 7B Experience Agent was fine-tuned with a lower learning rate (2×10^{-4}) and targeted

only the four attention projection layers (query, key, value, output), consistent with Gemma's architectural design, with a maximum sequence length of 768 tokens to accommodate multi-role work history narratives. All three agents used LoRA rank 32, $\alpha = 64$, dropout 0.05, cosine learning rate scheduling, 5 training epochs, and 4-bit NF4 quantisation with an effective batch size of 16.

3) INTERPRETATION OF RESULTS

All ten top-ranked configurations assign fine-tuned Gemma 7B to the Experience Agent, with the sole exception of rank 9 (Gemma + Qwen + Llama pre-trained; $R^2 = 0.7167$). The top five configurations use exclusively fine-tuned models across all agent roles. Ranks 6 and 7 each introduce one pre-trained Mistral agent in the Skills role ($R^2 = 0.7352$ and 0.7341 respectively), and rank 8 introduces pre-trained Mistral as the Education agent ($R^2 = 0.7282$), each step yielding a further decline relative to all-fine-tuned configurations. Rank 10 (Llama pre-trained Education, fine-tuned Mistral Skills, fine-tuned Gemma Experience; $R^2 = 0.7132$) achieves a notably high Pearson r of 0.9056, illustrating that strong linear correlation does not guarantee high R^2 : a systematic score offset from the pre-trained Llama agent raises residual variance despite strong rank-ordering ability.

Overall, these results suggest that *fine-tuned model quality across all agent roles* is the primary determinant of heterogeneous multi-agent performance: the five highest-ranked configurations use only fine-tuned models throughout, and introducing even a single pre-trained agent reduces R^2

TABLE 6. Top-10 heterogeneous multi-agent configurations by R^2 against the average annotator score. *Edu*, *Ski*, and *Exp* denote the model assigned to the Education, Skills, and Experience agents respectively. The best overall configuration is highlighted in bold.

Edu	Ski	Exp	MAE	MSE	RMSE	r	R^2
Qwen (FT)	Qwen (FT)	Gemma (FT)	0.0654	0.00635	0.0797	0.8959	0.7664
Mistral (FT)	Qwen (FT)	Gemma (FT)	0.0650	0.00640	0.0800	0.8935	0.7644
Gemma (FT)	Qwen (FT)	Gemma (FT)	0.0589	0.00682	0.0826	0.8680	0.7492
Qwen (FT)	Mistral (FT)	Gemma (FT)	0.0643	0.00691	0.0831	0.8785	0.7458
Mistral (FT)	Mistral (FT)	Gemma (FT)	0.0634	0.00699	0.0836	0.8750	0.7428
Mistral (FT)	Mist.(pre)	Gemma (FT)	0.0633	0.00720	0.0849	0.8741	0.7352
Qwen (FT)	Mist.(pre)	Gemma (FT)	0.0644	0.00723	0.0850	0.8773	0.7341
Mist.(pre)	Gemma (FT)	Gemma (FT)	0.0675	0.00739	0.0860	0.8849	0.7282
Gemma (FT)	Qwen (FT)	Llama (pre)	0.0697	0.00770	0.0878	0.8639	0.7167
Llama (pre)	Mistral (FT)	Gemma (FT)	0.0733	0.00780	0.0883	0.9056	0.7132

(FT) = fine-tuned; (pre) = pre-trained zero-shot

noticeably. Future heterogeneous multi-agent systems should fine-tune each component model independently on its designated input before combination, rather than relying on pre-trained agents for any role.

G. COMPARISON WITH STATE-OF-THE-ART COMMERCIAL MODELS

Table 7 reports zero-shot performance of five leading commercial models against all three annotator targets. Among the commercial baselines, Claude Sonnet 4.5 achieved the best overall performance with MAE = 0.0972, Pearson r = 0.6256, and R^2 = 0.3360 against the average annotator score. Gemini 3 followed with MAE = 0.1278, r = 0.5933, and R^2 = 0.2020. Grok (Auto, xAI) achieved MAE = 0.1208, r = 0.6050, but R^2 = -0.1959, indicating that while its predictions are linearly correlated with human scores, they are systematically biased. Mistral Large produced the lowest correlation (r = 0.0197) and a negative R^2 (-0.3260) against the average target, performing only marginally above chance. GPT-5 exhibited the poorest overall performance among the five commercial models, MAE = 0.2520, r = 0.1133, and R^2 = -2.1134 against the average annotator score, suggesting substantial over-scoring bias under zero-shot conditions on this dataset.

Comparing commercial baselines against our proposed systems, the best fine-tuned single-agent model (Qwen 2.5-14B augmented, R^2 = 0.7294) outperforms the strongest commercial baseline (Claude Sonnet 4.5, R^2 = 0.3360) by 0.3934 R^2 units, despite operating at a fraction of the parameter count and inference cost. Similarly, the best heterogeneous multi-agent configuration (R^2 = 0.7664, Qwen + Qwen + Gemma) surpasses all commercial baselines by a substantial margin. Even the few-shot Mistral Small 22B configuration (MAE = 0.0824, r = 0.6856) exceeds all commercial zero-shot baselines except Claude in Pearson r , demonstrating that domain-specific prompting with open-source models is competitive with proprietary systems at zero cost. This confirms that fine-tuning on the CareerCorpus dataset, even with a limited augmented training set, provides substantial and consistent advantages over state-of-the-art commercial systems evaluated in zero-shot settings.

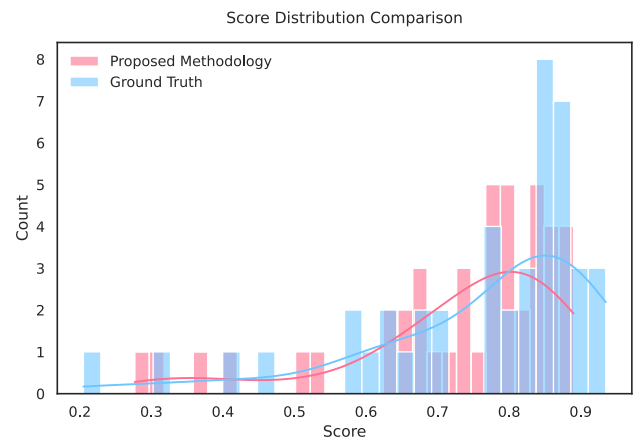


FIGURE 4. Score distribution comparison between the proposed methodology (Qwen_{FT}+Qwen_{FT}+Gemma_{FT}) and the ground-truth annotator average on the held-out test set.

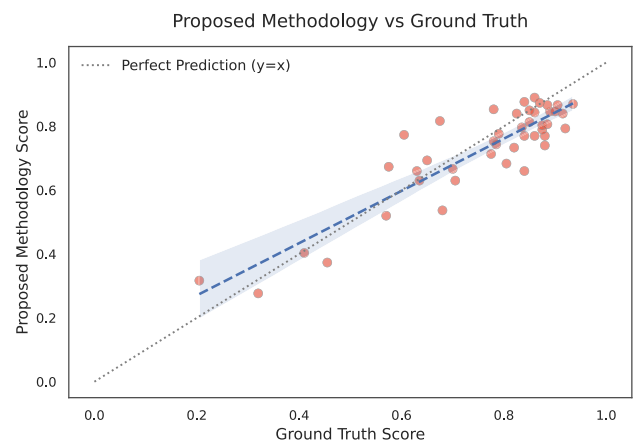


FIGURE 5. Predicted vs. ground-truth scores for the best heterogeneous multi-agent configuration on the held-out test set (Pearson r = 0.8959, MAE = 0.0654). The dashed line denotes perfect prediction ($y = x$).

H. QUANTITATIVE PERFORMANCE OF THE PROPOSED METHODOLOGY

Fig. 4 presents the score distribution of the best heterogeneous multi-agent configuration (Qwen_{FT}+Qwen_{FT}+Gemma_{FT}) compared with the ground-truth annotator



FIGURE 6. LIME explainability plots for the best heterogeneous multi-agent configuration ($Qwen_{F1} + Qwen_{F1} + Gemma_{F1}$) across all six job domains (rows) and three evaluation agents (columns). Green bars indicate tokens that positively contribute to the predicted score; red bars indicate negative contributions.

average on the 44-resume held-out test set. The predicted distribution closely mirrors the ground-truth distribution in the high-score range (0.8–0.9), confirming that the model captures the overall scoring tendency of expert annotators. Minor discrepancies at the lower end of the scale (below 0.6) reflect the relative scarcity of low-scoring resumes in the stratified test partition, which limits calibration for that region.

The scatter plot of individual forecasts against ground-truth scores is displayed in Fig. 5. Strong predictive accuracy is confirmed by the close clustering around the $y = x$ reference line (Pearson $r = 0.8959$, MAE = 0.0654). The few outlier points are associated with mid-range resumes (ground truth

0.4–0.6), where greater residuals are produced by the low training coverage.

I. QUALITATIVE CASE STUDY WITH COMPONENT TEXTS

Table 8 shows the actual text for each resume component along with the agent scores, the aggregated agent score, and the human annotator average. This lets readers see exactly what the candidate’s credentials and achievements are, while comparing them to agent predictions.

J. LIME-BASED EXPLAINABILITY ANALYSIS

To provide token-level transparency into each agent’s scoring decisions, we applied LIME (Local Interpretable

TABLE 7. Zero-shot performance of commercial SOTA models across all three evaluation targets, with full regression metrics. Negative R^2 values indicate predictions worse than a constant mean baseline.

Model	Annotator-1					Annotator-2					Average				
	MAE	MSE	RMSE	r	R^2	MAE	MSE	RMSE	r	R^2	MAE	MSE	RMSE	r	R^2
Claude Sonnet 4.5	0.1168	0.0287	0.1693	0.6387	0.2385	0.1052	0.0178	0.1334	0.5261	0.2744	0.0972	0.0181	0.1344	0.6256	0.3360
GPT-5	0.2486	0.0844	0.2905	0.0224	-1.2423	0.2737	0.0952	0.3086	0.2109	-2.8839	0.2520	0.0846	0.2909	0.1133	-2.1134
Gemini 3	0.1368	0.0262	0.1620	0.5862	0.3028	0.1484	0.0275	0.1658	0.5233	-0.1216	0.1278	0.0217	0.1473	0.5933	0.2020
Grok (Auto, xAI)	0.1500	0.0504	0.2245	0.6162	-0.3385	0.1039	0.0250	0.1581	0.5106	-0.0191	0.1208	0.0325	0.1803	0.6050	-0.1959
Mistral Large	0.1491	0.0531	0.2305	0.1017	-0.4118	0.1161	0.0293	0.1712	-0.0846	-0.1952	0.1226	0.0360	0.1899	0.0197	-0.3260
Proposed systems (best per category, Average target only)															
Best Single-Agent (Ours)	0.0847	0.0126	0.1124	0.8552	0.6645	0.0807	0.0124	0.1115	0.8221	0.4928	0.0644	0.0074	0.0858	0.8541	0.7294
Best Homogeneous MA (Ours)	0.0793	0.0116	0.1079	0.8478	0.6909	0.1109	0.0171	0.1308	0.7893	0.3024	0.0775	0.0092	0.0959	0.8736	0.6619
Best Heterogeneous MA (Ours)	0.0723	0.0097	0.0982	0.8673	0.7434	0.0957	0.0134	0.1157	0.8121	0.4536	0.0654	0.0064	0.0797	0.8959	0.7664

Note: All commercial models evaluated via official web interfaces in zero-shot setting using default generation parameters.

TABLE 8. Component-level breakdown of a research assistant resume (agent predictions vs annotator averages).

Domain	Component & Text	Agent & Score	Final Agent Score	Annotator Avg
Research Assistant	Education M.Sc. in CSE (CGPA 3.97/4.0) & B.Sc. in CSE (CGPA 3.24/4.0), North South University	Education Agent: 0.84	0.80	0.92
	Skills Skilled in Cybersecurity, Cloud Computing (OCI), DevOps, Linux, NS-3, HELICS, and Blockchain. Certified Ethical Hacker (Master, CEH). Published 6 papers incl. IEEE ICCCNT 2024, ICSGSC 2024, Springer 2024	Skill Agent: 0.89		
	Experience Asst. Manager (Cloud), BDCCL (2024–); Grad RA, NSU (2022–); RA, UIU (2024–); GTA, NSU (2023–24); Lab Instructor, Ethics Advance Tech (2023); Solution Engr., OneWorld InfoTech (2020–21). Cybersecurity R&D	Experience Agent: 0.67		

Each component shows the agent type and predicted score. Final Agent Score is the mean of the three component scores. Annotator Avg is the average of human Annotator-1 and Annotator-2 scores.

Model-agnostic Explanations) [37] to the fine-tuned models of the best heterogeneous configuration. For each domain-agent combination, LIME perturbs the input text and fits a locally linear surrogate model to identify which tokens most strongly increase (positive contribution, shown in green) or decrease (negative contribution, shown in red) the predicted quality score. This analysis was conducted across all six job domains and all three evaluation dimensions (Education, Skills, Experience) on representative test-set resumes.

Fig. 6 presents the LIME explanation plots. Each row corresponds to one job domain; each column corresponds to one agent. Several domain-specific patterns emerge. For the *Accountant* domain, the Education Agent assigns high positive weight to degree-level tokens such as *B.Com* and *MBA*, while the Skills Agent responds strongly to accounting-specific tools (*Tally*, *QuickBooks*). In the *Research Assistant* domain, the Education Agent heavily weights postgraduate indicators (*M.Sc.*, *CGPA*) and the Skills Agent highlights publication-related tokens (*IEEE*, *published*). The Experience Agent has the highest sensitivity to seniority-level signals across all domains, including role progression keywords and years of service.

These findings provide auditable proof that the framework's component-level scores represent significant qualifying signals rather than merely keyword matching, confirming that each refined agent learns domain-specific feature representations in line with its assessment focus.

K. COMPUTATIONAL COMPLEXITY ANALYSIS

Table 9 reports the per-agent computational complexity of the best heterogeneous multi-agent configuration (Qwen_{FT}+Qwen_{FT}+Gemma_{FT}) at inference. Total and LoRA adapter parameter counts were obtained directly from named model tensors; adapter sizes correspond to saved `.safetensors` weights on disk; FLOPs were estimated using the standard NLP approximation described in the table caption; inference time is the mean of eight forward passes after two warm-up runs; and peak GPU memory reflects the maximum allocated memory during a single forward pass.

A key finding is that QLoRA fine-tuning updates only a small fraction of each model's total parameters: 0.97% for both Qwen-based agents (137.63M of 14,129.10M) and 0.30% for the Gemma-based Experience Agent (25.69M of 8,563.38M), while the remaining base model weights remain completely frozen. This confirms the parameter efficiency of the QLoRA approach adopted in this work. The Education Agent (seq_len = 512) requires fewer FLOPs (14.47 T) and lower inference time (0.81 s) than the Skills Agent (seq_len = 768; 21.70 T; 1.22 s), consistent with the theoretical linear scaling of FLOPs with sequence length. The Gemma-based Experience Agent is the most lightweight of the three, requiring only 13.15 T FLOPs, 0.57 s per inference, and 8.11 GB peak GPU memory — comfortably within the P100's 16 GB HBM2 budget. All three agents fit on a

TABLE 9. Per-agent computational complexity of the best heterogeneous multi-agent configuration (Qwen_{FT}+Qwen_{FT}+Gemma_{FT}) at inference on an NVIDIA Tesla P100 GPU (16 GB HBM2). FLOPs estimated as $2 \times N_{\text{params}} \times L_{\text{seq}}$.

	Education Agent (Qwen 2.5-14B)	Skills Agent (Qwen 2.5-14B)	Experience Agent (Gemma 7B)
Metric			
Total Parameters (M)	14,129.10	14,129.10	8,563.38
LoRA Parameters (M)	137.63	137.63	25.69
LoRA % of Total	0.97%	0.97%	0.30%
Adapter Size (MB)	525.11	525.11	98.04
Max Sequence Length	512	768	768
FLOPs (T)	14.47	21.70	13.15
Inference Time (s)	0.81	1.22	0.57
Peak GPU Memory (GB)	13.17	13.27	8.11

single P100 GPU, and since they operate on disjoint resume sections they can be executed in parallel, making the effective wall-clock inference time equal to the slowest agent (1.22 s for the Skills Agent).

V. CONCLUSION

We introduce an automated resume evaluation framework based on the CareerCorpus dataset, comprising 302 annotated resumes across six job domains and three seniority levels. Our system leverages parameter-efficient LoRA fine-tuning and a multi-agent architecture to assess Education, Skills, and Experience in a structured and interpretable manner. Across five modeling paradigms—from TF-IDF baselines ($R^2 \approx 0.64$) through pre-trained transformers ($R^2 = 0.75$) to fine-tuned LLMs—each step yielded measurable gains, with the heterogeneous multi-agent configuration (Qwen_{FT}+Qwen_{FT}+Gemma_{FT}) achieving the best overall performance ($R^2 = 0.7664$, MAE = 0.0654) and outperforming all five commercial baselines by a substantial margin. Data augmentation proved essential for stable fine-tuning on limited data, and task-specific model assignment across agent roles emerged as the dominant performance driver. The full pipeline requires 4 hours 32 minutes of fine-tuning on a single NVIDIA Tesla P100 GPU, with a maximum inference time of 1.22 seconds per resume at a peak GPU memory footprint of 13.27 GB. Expanding training data, cross-domain generalization, and including structured resume elements are among the upcoming projects. This effort provides a replicable foundation and a useful tool for automated resume review in real-world employment systems through the release of CareerCorpus and a modular multi-agent codebase.

ETHICS STATEMENT

Every experiment in this study was carried out in compliance with accepted ethical guidelines. A publicly accessible, authorized source provided the CareerCorpus dataset used in this study [21]. No new data collection involving human participants was performed as part of this research. The models developed in this study are intended solely

for academic research purposes. The authors caution that automated resume evaluation systems should not be deployed in real-world hiring processes without appropriate human oversight, as such systems may perpetuate or amplify biases present in training data.

DATA AVAILABILITY

The CareerCorpus dataset used in this study is publicly available at <https://data.mendeley.com/datasets/wzzwn37gmd/1>

CODE AVAILABILITY

The full experimental codebase, including fine-tuning scripts, multi-agent evaluation pipelines, and prompt templates, is available at <https://github.com/AdibAFC/Resume-Screening>

REFERENCES

- [1] Society for Human Resource Management (SHRM). (Apr. 11, 2022). *The Real Costs of Recruitment*. [Online]. Available: <https://www.shrm.org/>
- [2] Glassdoor. (Aug. 9, 2017). *How Long Does it Take to Hire? Interview Duration in 25 Countries*. [Online]. Available: <https://www.glassdoor.com/>
- [3] J. S. Black and P. van Esch, "AI-enabled recruiting: What is it and how should a manager use it?" *Bus. Horizons*, vol. 63, no. 2, pp. 215–226, Mar. 2020, doi: [10.1016/j.bushor.2019.12.001](https://doi.org/10.1016/j.bushor.2019.12.001).
- [4] M. Raghavan, S. Barocas, J. Kleinberg, and K. Levy, "Mitigating bias in algorithmic hiring: Evaluating claims and practices," in *Proc. ACM Conf. Fairness, Accountability, Transparency (FAccT)*, Barcelona, Spain, 2020, pp. 469–481, doi: [10.1145/3351095.3372828](https://doi.org/10.1145/3351095.3372828).
- [5] P. K. Roy, S. S. Chowdhury, and R. Bhatia, "A machine learning approach for automation of resume recommendation system," *Proc. Comput. Sci.*, vol. 167, pp. 2318–2327, Jan. 2020, doi: [10.1016/j.procs.2020.03.284](https://doi.org/10.1016/j.procs.2020.03.284).
- [6] A. Alsharef, H. Nassour, and J. Sharma, "Exploring the efficiency of text-similarity measures in automated resume screening for recruitment," in *Proc. 10th Int. Conf. Comput. Sustain. Global Develop. (INDIACom)*, New Delhi, India, Mar. 2023, pp. 36–42.
- [7] P. Singla, J. Kaur, Anju, A. Soni, A. Tuteja, and S. Sharma, "Streamlining talent acquisition: A machine learning approach to automated resume screening," in *Proc. 2nd Int. Conf. Adv. Comput. Commun. Technol. (ICACCTech)*, Sonapat, India, Nov. 2024, pp. 69–75, doi: [10.1109/ICACCTECH65084.2024.00022](https://doi.org/10.1109/ICACCTECH65084.2024.00022).
- [8] A. B. Nandrajog, A. Jain, A. Sharma, Deepa, and L. Singla, "Intelligent resume screening and candidate ranking using CNN-GRU: A deep learning approach for smarter recruitment," in *Proc. 5th Int. Conf. Intell. Technol. (CONIT)*, Hubballi, India, Jun. 2025, pp. 1–4, doi: [10.1109/CONIT65521.2025.11167664](https://doi.org/10.1109/CONIT65521.2025.11167664).
- [9] A. Deshmukh and A. Raut, "Enhanced resume screening for smart hiring using sentence-bidirectional encoder representations from transformers (S-BERT)," *Int. J. Adv. Comput. Sci. Appl.*, vol. 15, no. 8, pp. 241–247, 2024, doi: [10.14569/ijacsa.2024.0150825](https://doi.org/10.14569/ijacsa.2024.0150825).
- [10] E. Salakar, J. Rai, A. Salian, Y. Shah, and J. Wadmare, "Resume screening using large language models," in *Proc. 6th Int. Conf. Adv. Sci. Technol. (ICAST)*, Mumbai, India, Dec. 2023, pp. 494–499, doi: [10.1109/icast59062.2023.10454984](https://doi.org/10.1109/icast59062.2023.10454984).
- [11] C. Gan, Q. Zhang, and T. Mori, "Application of LLM agents in recruitment: A novel framework for resume screening," 2024, *arXiv:2401.08315*.
- [12] F. P.-W. Lo, J. Qiu, Z. Wang, H. Yu, Y. Chen, G. Zhang, and B. Lo, "AI hiring with LLMs: A context-aware and explainable multi-agent framework for resume screening," 2025, *arXiv:2504.02870*.
- [13] S. Bian, X. Chen, W. X. Zhao, K. Zhou, Y. Hou, Y. Song, T. Zhang, and J.-R. Wen, "Learning to match jobs with resumes from sparse interaction data using multi-view co-teaching network," in *Proc. 29th ACM Int. Conf. Inf. Knowl. Manage.*, Oct. 2020, pp. 65–74.
- [14] C. Qin, H. Zhu, T. Xu, C. Zhu, L. Jiang, E. Chen, and H. Xiong, "Enhancing person-job fit for talent recruitment: An ability-aware neural network approach," in *Proc. 41st Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, Ann Arbor, MI, USA, Jun. 2018, pp. 25–34, doi: [10.1145/3209978.3210025](https://doi.org/10.1145/3209978.3210025).
- [15] A. Lambrecht and C. Tucker, "Algorithmic bias? An empirical study of apparent gender-based discrimination in the display of STEM career ads," *Manage. Sci.*, vol. 65, no. 7, pp. 2966–2981, Jul. 2019, doi: [10.1287/mnsc.2018.3093](https://doi.org/10.1287/mnsc.2018.3093).
- [16] D. F. Mujtaba and N. R. Mahapatra, "Fairness in AI-driven recruitment: Challenges, metrics, methods, and future directions," 2024, *arXiv:2405.19699*.
- [17] H. Mu, "AI security beyond core domains: Resume screening as a case study of adversarial vulnerabilities in specialized LLM applications," 2025, *arXiv:2512.20164*.
- [18] F. Wu, A. Ye, Z. Cai, and J. Chen, "Towards bias-free recruitment: Adversarial contrastive tuning for LLM-based resume screening," *Inf. Softw. Technol.*, vol. 193, May 2026, Art. no. 108040, doi: [10.1016/j.infsof.2026.108040](https://doi.org/10.1016/j.infsof.2026.108040).
- [19] A. Aka, E. Palikot, A. Ansari, and N. Yazdani, "Better together: Quantifying the benefits of AI-assisted recruitment," 2025, *arXiv:2507.08029*.
- [20] A. Heakl, "ResumeAtlas: Revisiting resume classification with large-scale datasets and large language models," in *Proc. 6th Int. Conf. AI Comput. Linguistics*, 2024.
- [21] M. S. Chowdhury and A. F. Chowdhury. (2026). *CareerCorpus: A Professionally Annotated Multi-Domain Resume Dataset for Automated Evaluation*. Mendeley Data. [Online]. Available: <https://data.mendeley.com/datasets/wzzwn37gmd/1>
- [22] K. Yu, G. Guan, and M. Zhou, "Resume information extraction with cascaded hybrid model," in *Proc. 43rd Annu. Meeting Assoc. Comput. Linguistics (ACL)*, Ann Arbor, MI, USA, 2005, pp. 499–506.
- [23] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, Minneapolis, MN, USA, 2019, pp. 4171–4186.
- [24] Q. Chen, D. Kong, Y. Zhu, Z. Shen, C. Lu, Y. Li, and L. Zhang, "CareerMiner: Automatic extraction of professional network from large Chinese resume data," *Franklin Open*, vol. 6, Mar. 2024, Art. no. 100065, doi: [10.1016/j.fraope.2023.100065](https://doi.org/10.1016/j.fraope.2023.100065).
- [25] M. E. Erdem, "Automatic resume screening with content matching," in *Proc. 8th Int. Conf. Comput. Sci. Eng. (UBMK)*, Burdur, Turkey, Sep. 2023, pp. 554–558, doi: [10.1109/UBMK59864.2023.10286578](https://doi.org/10.1109/UBMK59864.2023.10286578).
- [26] C. S. Surya, G. Kiriti, K. S. Saketh, P. S. Charan, and A. Anjali, "Optimizing job matching through automated resume screening with NLP and machine learning," in *Proc. IEEE Int. Conf. Intell. Signal Process. Effective Commun. Technol. (INSPECT)*, Gwalior, India, Dec. 2024, pp. 1–6, doi: [10.1109/INSPECT63485.2024.10896014](https://doi.org/10.1109/INSPECT63485.2024.10896014).
- [27] M. A. Tejaswini, "Automated resume screening using nlp and machine learning," *Int. Sci. J. Eng. Manage.*, vol. 4, no. 5, pp. 1–7, May 2025, doi: [10.55041/isjem03751](https://doi.org/10.55041/isjem03751).
- [28] V. Khatri, V. Kumar, K. Thakur, V. Kumar, Tanisha, Vani, and R. Yadav, "AI-powered automated resume screening and job matching system using NLP and machine learning," *Int. J. Res. Eng. Sci. Manage.*, vol. 8, no. 10, pp. 58–62, Oct. 2025, doi: [10.65138/ijresm.v8i10.3367](https://doi.org/10.65138/ijresm.v8i10.3367).
- [29] R. Le, W. Hu, Y. Song, T. Zhang, D. Zhao, and R. Yan, "Towards effective and interpretable person-job fitting," in *Proc. 28th ACM Int. Conf. Inf. Knowl. Manage.*, Nov. 2019, pp. 1883–1892.
- [30] B. Hui, "Qwen2.5 technical report," 2024, *arXiv:2412.15115*.
- [31] H. T. Llama, "Llama 3: Open and efficient foundation language models," 2024, *arXiv:2407.21783*.
- [32] E. J. Hu, "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022.
- [33] T. Dettmers, A. Holtzman, A. Pagnoni, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in *Proc. Adv. Neural Inf. Process. Syst.*, New Orleans, LA, USA, 2023, pp. 10088–10115.
- [34] "Mistral small 3," Mistral AI, Paris, France, Tech. Rep., 2024.
- [35] T. Mesnard, "Gemma: Open models based on Gemini research and technology," 2024, *arXiv:2403.08295*.
- [36] J. Wei and K. Zou, "EDA: Easy data augmentation techniques for boosting performance on text classification tasks," in *Proc. Conf. Empirical Methods Natural Lang. Process. 9th Int. Joint Conf. Natural Lang. Process. (EMNLP-IJCNLP)*, Hong Kong, 2019, pp. 6381–6387.
- [37] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you?: Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, San Francisco, CA, USA, Aug. 2016, pp. 1135–1144.
- [38] S. Agarwal, "GPT-OSS-120b & GPT-OSS-20b model card," 2025, *arXiv:2508.10925*.



MD. SAGOR CHOWDHURY is currently pursuing the B.S. degree with the Department of Computer Science and Engineering, Chittagong University of Engineering and Technology (CUET), Chittagong, Bangladesh. His research interests include natural language processing, large language models, and automated recruitment systems.



ADIBA FAIROOZ CHOWDHURY is currently pursuing the B.S. degree with the Department of Computer Science and Engineering, Chittagong University of Engineering and Technology (CUET), Chittagong, Bangladesh. Her research interests include machine learning, NLP, and AI-driven human resource management systems.



AYESHA BANU is currently a Faculty Member with the Department of Computer Science and Engineering, Chittagong University of Engineering and Technology (CUET), Chittagong, Bangladesh. Her research interests include machine learning, deep learning, natural language processing, and artificial intelligence applications in software engineering and intelligent systems. Her recent work focuses on applying data-driven techniques to problems, such as social media

analysis, automated decision support, and computer vision.



RIAD HOSSAIN is currently a Faculty Member with the School of Science, Engineering and Technology, East Delta University, Chittagong, Bangladesh. His research interests include machine learning, natural language processing, explainable artificial intelligence, cybersecurity, and AI applications in healthcare, education, and recruitment. His recent work focuses on developing data-driven solutions for low-resource languages, particularly Bengali, and applying machine learning techniques to real-world problems, such as social media analysis, disease detection from speech signals, and intelligent decision-support systems.



MAHFUZULHOQ CHOWDHURY received the Ph.D. degree from INRS, University of Quebec, Montreal, Canada, in 2018. He is currently a Faculty Member and a Researcher with the Computer Science and Engineering Department, Chittagong University of Engineering and Technology. He also studied and worked with McGill University during his doctoral period. His major research interests include machine learning, LLM, deep learning, 6G, cloud computing, IoT, cyber security, image and video processing, and mobile app development. He has published several journals and conference papers in reputable publishers like Springer and IEEE.

...